

Qiskit Installation & Setup Guide (Steps 1–9)

Step 1 — Python & Jupyter (Quick Check)

1. Check your Python version:

```
python --version
```

2. If Python 3.12 is not installed, download it from:

<https://www.python.org/downloads/release/python-3129/>

During installation:

- Check “Add Python to PATH”
- Allow long path support if prompted

3. Launch Jupyter Notebook:

```
python -m notebook
```

A browser window should open with the Jupyter interface.

Step 2 — Understanding Qiskit Components

Qiskit is modular. We will explicitly install:

- qiskit (core framework)
- qiskit-aer (simulator backend)
- visualization libraries (for circuits and plots)

Step 3 — Install Qiskit Core

Run the following command inside a Jupyter Notebook cell:

```
!python -m pip install qiskit
```

This installs:

- QuantumCircuit
- gates and registers
- quantum information tools

Step 4 — Install the Simulator (Qiskit-Aer)

Run inside Jupyter:

```
!python -m pip install qiskit-aer
```

This installs the high-performance simulator used throughout the course.

Step 5 — Install Visualization Dependencies

These libraries are required to draw circuits and histograms:

```
!python -m pip install matplotlib pylatexenc qiskit[visualization]
```

They enable:

- circuit diagrams
- measurement histograms
- LaTeX-style gate labels

Step 6 — Restart the Kernel

After installing packages:

- In Jupyter, click Kernel → Restart Kernel → Restart

This step is mandatory so newly installed libraries are recognized.

Step 7 — Verify Qiskit Installation

Run:

```
import qiskit
print(qiskit.__version__)
```

A version number confirms successful installation.

Step 8 — First Quantum Circuit (Superposition)

Run the following code:

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

qc = QuantumCircuit(1)
qc.h(0)
qc.measure_all()

display(qc.draw("mpl"))
```

```
sim = AerSimulator()
result = sim.run(qc, shots=1000).result()
counts = result.get_counts()

print("Measurement results (1000 shots):")
print(counts)

plot_histogram(counts)
plt.show()
```

Step 9 — Expected Outcome

You should see:

- A drawn quantum circuit with a Hadamard gate
- Measurement counts close to 50% $|0\rangle$ and 50% $|1\rangle$
- A histogram visualizing the results

This completes the Qiskit setup required for the course.